File permissions in Linux

Project description

This project involved reviewing and managing file and directory permissions in a Linux environment to ensure that users within an organization's research team had appropriate levels of access. Using Linux commands such as `ls -la` and `chmod`, I examined existing permissions, identified unauthorized access, and modified permissions to align with the organization's security requirements. The objective was to protect sensitive research files, prevent unauthorized modifications, and ensure that access was restricted to authorized users.

Check file and directory details

```
researcher2@d64b73bfd0e1:~$ cd /home/researcher2/projects
researcher2@d64b73bfd0e1:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Nov  5 00:51 .
drwxr-xr-x 3 researcher2 research_team 4096 Nov  5 01:10 ..
-rw--w---- 1 researcher2 research_team   46 Nov  5 00:51 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Nov  5 00:51 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Nov  5 00:51 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Nov  5 00:51 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Nov  5 00:51 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Nov  5 00:51 project_t.txt
researcher2@d64b73bfd0e1:~/projects$
```

Describe the permissions string

Strings that start with a “d” are directories, strings that start with a “-” are files.
2nd, 3rd and 4th characters indicate user permissions: read (r), write (w), execute (x)
5th, 6th and 7th characters indicate group permissions
8th, 9th and 10th characters indicate all other users permissions

Change file permissions

```
researcher2@d64b73bfd0e1:~/projects$ chmod o-w project_k.txt
researcher2@d64b73bfd0e1:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Nov  5 00:51 .
drwxr-xr-x 3 researcher2 research_team 4096 Nov  5 01:10 ..
-rw--w---- 1 researcher2 research_team   46 Nov  5 00:51 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Nov  5 00:51 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Nov  5 00:51 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Nov  5 00:51 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Nov  5 00:51 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Nov  5 00:51 project_t.txt
researcher2@d64b73bfd0e1:~/projects$
```

Change file permissions on a hidden file

```
researcher2@d64b73bfd0e1:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@d64b73bfd0e1:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Nov  5 00:51 .
drwxr-xr-x 3 researcher2 research_team 4096 Nov  5 01:10 ..
-r--r----- 1 researcher2 research_team   46 Nov  5 00:51 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Nov  5 00:51 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Nov  5 00:51 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Nov  5 00:51 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Nov  5 00:51 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Nov  5 00:51 project_t.txt
researcher2@d64b73bfd0e1:~/projects$
```

Change directory permissions

```
researcher2@d64b73bfd0e1:~/projects$ chmod g-x drafts
researcher2@d64b73bfd0e1:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Nov  5 00:51 .
drwxr-xr-x 3 researcher2 research_team 4096 Nov  5 01:10 ..
-r--r----- 1 researcher2 research_team   46 Nov  5 00:51 .project_x.txt
drwx------ 2 researcher2 research_team 4096 Nov  5 00:51 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Nov  5 00:51 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Nov  5 00:51 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Nov  5 00:51 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Nov  5 00:51 project_t.txt
researcher2@d64b73bfd0e1:~/projects$
```

Summary

In this project, I examined and modified Linux file and directory permissions to improve access control within an organization's research environment. I used `ls -la` to inspect existing permissions, identify hidden files, and interpret the 10-character permission strings. I applied `chmod` to remove unauthorized write access, secure an archived hidden file, and restrict access to the drafts directory to its designated owner. These tasks strengthened my practical understanding of Linux permissions, file security, and the principle of least privilege, which are essential for protecting organizational information and maintaining secure systems.